# Thornfield Analytic Guild

Fictional laboratory report — for software evaluation only

| | |
|---|---|
| Patient: Aarav Mehta | Age/Sex: 42 Y / Male |
| Sample ID: SYN-0006 | Report Date: 15/06/2024 |
| Collected: 15-Jun-2024 | Status: Final (synthetic) |

## COMPLETE BLOOD COUNT

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| Hemoglobin | 14.2 | | g/dL | 13.0 - 17.0 |
| RBC Count | 4.85 | | mill/cumm | 4.5 - 5.5 |
| Total Leucocyte Count | 7200 | | /cumm | 4000 - 11000 |
| Platelet Count | 2.45 | | Lakhs/cumm | 1.5 - 4.5 |
| PCV | 42.5 | | % | 40 - 50 |
| MCV | 87.6 | | fL | 83 - 101 |
| MCH | 29.3 | | pg | 27 - 32 |
| MCHC | 33.4 | | Gms% | 31.5 - 34.5 |

## DIFFERENTIAL COUNT

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| Neutrophils | 58 | | % | 40 - 70 |
| Lymphocytes | 32 | | % | 20 - 40 |
| Monocytes | 6 | | % | 2 - 10 |
| Eosinophils | 3 | | % | 1 - 6 |
| Basophils | 1 | | % | 0 - 2 |

## LIPID PROFILE

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| Cholesterol - Total | 198 | | mg/dL | Desirable < 200<br>Borderline 200-239<br>High >= 240 |
| LDL Cholesterol | 138 | H | mg/dL | < 130 |
| HDL Cholesterol | 48 | | mg/dL | > 40 |
| Triglycerides | 142 | | mg/dL | < 150 |
| VLDL | 28 | | mg/dL | 5 - 40 |

## LIVER FUNCTION

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| SGOT (AST) | 28 | | U/L | Up to 40 |
| SGPT (ALT) | 32 | | U/L | Up to 41 |
| Alkaline Phosphatase | 78 | | U/L | 40 - 129 |
| GGT | 24 | | U/L | < 55 |
| Bilirubin Total | 0.7 | | mg/dL | 0.3 - 1.2 |
| Albumin | 4.2 | | g/dL | 3.5 - 5.2 |

## KIDNEY FUNCTION

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| Blood Urea | 28 | | mg/dL | 17 - 43 |
| Serum Creatinine | 0.9 | | mg/dL | 0.7 - 1.3 |
| Uric Acid | 5.4 | | mg/dL | 3.5 - 7.2 |

## ELECTROLYTES

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| Sodium | 140 | | mEq/L | 136 - 145 |
| Potassium | 4.2 | | mEq/L | 3.5 - 5.1 |

## THYROID

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| TSH | 2.15 | | µIU/mL | 0.4 - 4.2 |
| Total T3 | 112 | | ng/dL | 80 - 200 |
| Total T4 | 7.8 | | µg/dL | 5.1 - 14.1 |
| Free T4 | 1.15 | | ng/dL | 0.8 - 1.8 |

## DIABETES

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| Fasting Blood Sugar | 98 | | mg/dL | 70 - 100 |
| HbA1c | 5.4 | | % | 4.0 - 5.6 |

## VITAMINS & IRON

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| 25-OH Vitamin D | 22.0 | L | ng/mL | 30 - 100 |
| Vitamin B12 | 420 | | pg/mL | 200 - 900 |
| Ferritin | 85 | | ng/mL | 30 - 400 |
| Serum Iron | 95 | | µg/dL | 60 - 170 |

## INFLAMMATION

| Test | Result | Flag | Unit | Normal Range |
|---|---|---|---|---|
| C-Reactive Protein | 4.5 | H | mg/L | < 3.0 |
| ESR | 12 | | mm/hr | 0 - 20 |

*** End of synthetic report ***